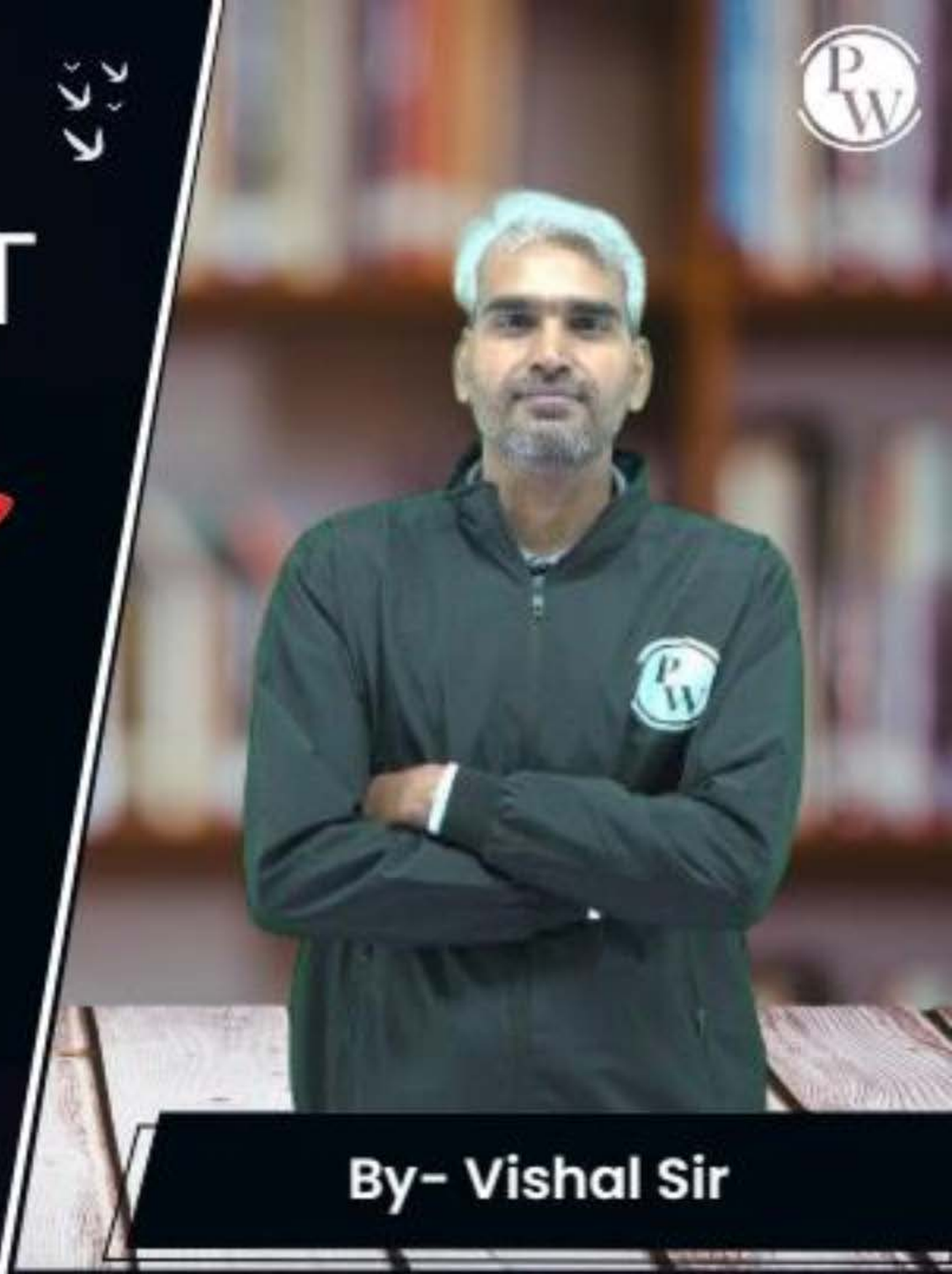
# Computer Science & IT



## Discrete Mathematics

## Mathematical Logic

**Lecture No. 02**



By- Vishal Sir

# Recap of Previous Lecture



**Topic** Propositions and their types

**Topic** Connectives

**Topic** Tautology, Contradiction, Contingency and Satisfiable propositional functions

Slide

# Topics to be Covered



Topic Logical implication

Topic Logical equivalence

Topic Argument and rules of inference

Slide



# Topic : Logical Implication / Implication

- Let P and Q are any two propositional functions.

- Whenever P is true if Q is also true then P logically implies Q.

  and if it is given that P logically implies Q, then Whenever P is true Q is guaranteed to be true

  P logically implies Q if and only if $P \rightarrow Q$ is a tautology.

$S_1$: "$P$ logically implies $Q$"

Meaning: $P \rightarrow Q$ is a tautology or Whenever $P$ is true $Q$ is guaranteed to be true

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Q:- Check whether "$P$ logically implies $Q$" is valid or invalid

- If $P \rightarrow Q$ is a tautology, then "$P$ logically implies $Q$" is valid
- If $P \rightarrow Q$ is not a tautology, then "$P$ logically implies $Q$" is invalid



# Topic : Logical Equivalence / Equivalence

- If P and Q are any two propositional functions, then P equivalent to Q is written as $P \equiv Q$. or $P \iff Q$

- P and Q are said to be equivalent if and only if they have same truth table.

- $P \equiv Q$ if and only if $P \leftrightarrow Q$ is a tautology.
- $P \equiv Q$ if and only if, $P \rightarrow Q$ is a tautology as well as $Q \rightarrow P$ is a tautology.
- $P \equiv Q$ if and only if, P logically implies Q as well as Q logically implies P



# Topic : Some important equivalences

① $\sim(\sim p) \equiv p$ } Double Complementation

② $p \vee q \equiv q \vee p$
$p \wedge q \equiv q \wedge p$ } Commutative

③ $(p \vee q) \vee r \equiv p \vee (q \vee r)$
$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ } Associative

$\wedge \equiv \cdot$

$\vee \equiv +$



# Topic : Some important equivalences

④ $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$

$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$

} Distributive Property

⑤ $\sim(P \wedge Q) \equiv (\sim P \vee \sim Q)$

$\sim(P \vee Q) \equiv (\sim P \wedge \sim Q)$

} De' Morgan law

⑥ $P \wedge (P \vee Q) \equiv P$

$P \vee (P \wedge Q) \equiv P$

} Absorption law



# Topic : Some important equivalences

(7)

$P \wedge P \equiv P$
$P \vee P \equiv P$ } Idempotent law

$P \vee T \equiv T$
$P \wedge F \equiv F$ } Domination law

$P \vee F \equiv P$
$P \wedge T \equiv P$ } Identity

| |
|---|
| $T \wedge T \equiv T$ |
| $T \vee T \equiv T$ |
| $T \wedge F \equiv F$ |
| $T \vee F \equiv T$ |
| $F \vee F \equiv F$ |
| $F \wedge F \equiv F$ |

- Some important Equivalences :-



$$P \rightarrow Q \equiv \sim P \vee Q$$

$$P \rightarrow Q \equiv \sim Q \rightarrow \sim P$$

$$P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$$

H.W

Q: Simplify the following statement

$$\{\sim p \wedge (\sim q \wedge r)\} \vee \{(q \wedge r) \vee (p \wedge r)\}$$

(a) True
(b) False
(c) p
(d) q
(e) r

H.W:

Q: Simplify the following Expression

$$[(p \vee q) \wedge \sim\{\sim p \wedge (\sim q \vee \sim r)\}] \vee [(\sim p \wedge \sim q) \vee (\sim p \wedge \sim r)]$$

H.W. Q: Which of the following is/are valid?

(a) $(P \wedge Q)$ logically implies $(P \leftrightarrow Q)$

(b) $(P \leftrightarrow Q)$ logically implies $(P \rightarrow Q)$

(c) $(P \leftrightarrow Q)$ logically implies $(P \rightarrow \sim Q)$

(d) $Q$ logically implies $(P \rightarrow Q)$

H.W. Q: Which of the following is/are tautology ?

(a) $(P \wedge Q) \longrightarrow (P \longleftrightarrow Q)$

(b) $(P \longleftrightarrow Q) \longrightarrow (P \longrightarrow Q)$

(c) $(P \longleftrightarrow Q) \longrightarrow (P \longrightarrow \sim Q)$

(d) $Q \longrightarrow (P \longrightarrow Q)$

H.W.

Q: Which of the following is/are tautology?

(a) $(a \vee b) \rightarrow (b \wedge c)$

(b) $(a \wedge b) \rightarrow (b \vee c)$

(c) $(a \vee b) \rightarrow (b \rightarrow c)$

(d) $(a \rightarrow b) \rightarrow (b \rightarrow c)$

LHS True then RHS True

or

R.H.S is false then L.H.S False

H.W

Q: Consider the following statements

- $S_1$: $\{(A \wedge B) \longrightarrow C\} \equiv \{(A \rightarrow C) \wedge (B \rightarrow C)\}$
- $S_2$: $\{(A \vee B) \longrightarrow C\} \equiv \{(A \rightarrow C) \vee (B \rightarrow C)\}$

then which of the following is true ?

(a) Both $S_1$ & $S_2$ are valid

(b) Only $S_1$ is valid

(c) Only $S_2$ is valid

(d) Neither $S_1$ nor $S_2$ is valid.

H.W. Q: Which of the following is are true ?

(a) $\{a \rightarrow (b \vee c)\} \equiv \{(a \wedge \sim b) \rightarrow c\}$

(b) $\{(P \rightarrow Q) \wedge (R \rightarrow Q)\} \equiv \{(P \vee R) \rightarrow Q\}$

(c) $\{(P \rightarrow Q) \vee (R \rightarrow Q)\} \equiv \{(P \wedge R) \rightarrow Q\}$

(d) All of the above

H.W.

Q. Consider the following propositional statement.

$$\{P \rightarrow (Q \vee R)\} \longrightarrow \{(P \wedge Q) \rightarrow R\}$$

Above propositional statement is.

- (a) Satisfiable but not valid {i.e. Contingency}
- (b) Valid
- (c) Contradiction

Statement $S_1$: If it rains today then Cricket match is not played

Statement $S_2$: Cricket match is not played

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Conclusion: It rained today

→ it is an argument

→ it may be valid
or
it may be invalid

Our job is to check whether the argument is valid or invalid



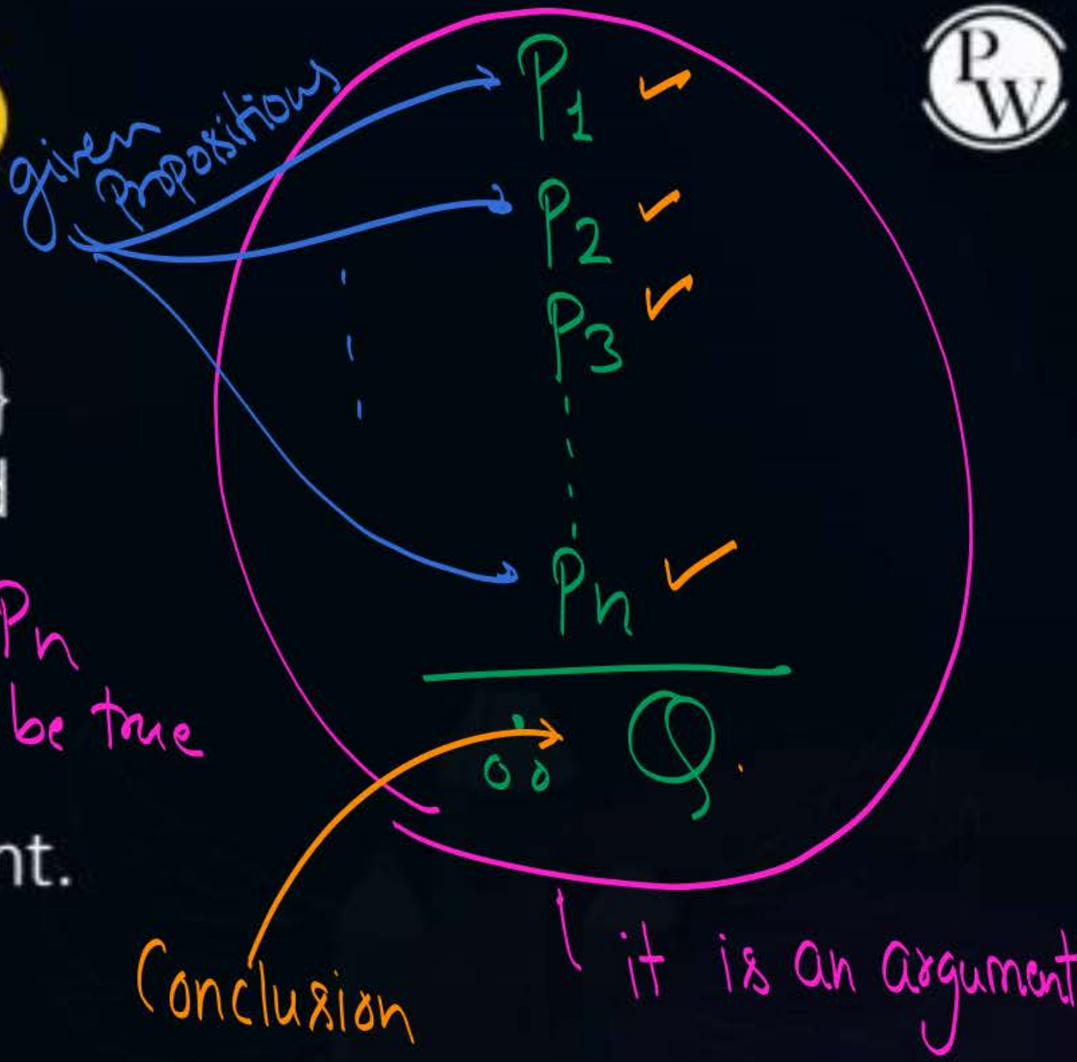
# Topic : Argument / Inference

The statement that,

- A set of premises { P1, P2, P3,.....,Pn} yields another proposition Q is called an argument.

- The argument is, that whenever $P_1, P_2, \ldots P_n$ are true then Conclusion Q is guaranteed to be true

Q is called conclusion of the argument.





# Topic : Argument / Inference

- Argument may be valid or invalid.
- The process of reasoning whether the argument is valid or invalid is called inference.
- If conclusion Q can be inferred from the set of premises by applying some rules of inference and equivalences, then argument is said to be valid otherwise invalid.



# Topic : Rules of inference

Any valid reasoning is rule of inference.

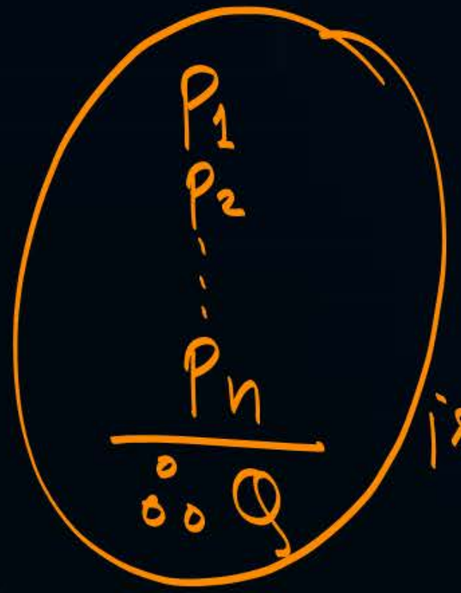


# Topic : Argument / Inference

Following statements are equivalent,

- Argument {P1, P2, P3,.....,Pn} $\vdash Q$ is valid. (yields)
- {P1, P2, P3,.....,Pn} Logically implies $Q$ is valid
- {P1 ∧ P2 ∧ P3 ∧ ..... ∧ Pn} → $Q$ is a tautology
- Conclusion Q follows from the premises $\{P_1, P_2, \cdots P_n\}$

$\equiv$




is valid



# 2 mins Summary

Topic Logical implication

Topic Logical equivalence

Topic Argument and rules of inference

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THANK - YOU
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